

Systematic Analysis Plan

FKB327 Injection Site Pain Data

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1 BACKGROUND

FKB327 is a biosimilar of the adalimumab reference product (RP). The formulation excipients of the FKB327 differ from those of the citrate-containing formulation of the RP, thus injection-site pain (ISP) at subcutaneous injection were assessed in single-dose studies (FKB327-001¹) and -004) in healthy subjects and multiple-dose studies (FKB327-002 double blind study (DBT) (NCT02260791)³) and -003 open label extension study (OLE) (NCT02405780)⁴) in patients with rheumatoid arthritis. Outline of individual study design can be referred to publications^{1,2}), trial registration^{3,4}), FKB327 EPAR Public assessment report⁵), and/or appendices 1 to 4 for studies 001 to 004.

In all 4 studies, ISP was assessed using visual analog scale (VAS) with a horizontal scale from 0 to 100 mm (0 to 100 scale in KB327-002 and -003 using ePRO device) with the left end point signifying “no pain” and the right end point signifying “intolerable pain”. In 001 and 004 studies by single-dose 001 and 004 studies, assessments of ISP were performed immediately after dosing and at 0.5, 1, 12, and 24 hours post dose in both studies. In the multiple-dose studies (studies 002 and 003), assessments of ISP were performed within 30 minutes of administration of the first dose of study drugs. In order to evaluate potential difference in ISP among products, devices of FKB327, or the other subjects’ background factors, this systemic anapysis using pooled data from these studies was planned.

2 DATA TO BE ANALYZED IN SYSTEMIC ANALYSIS

The VAS data of ISP from FKB327-001, -002, -003, and -004 are integrated for the systemic analysis. All cumulative data will be analyzed for the randomized subjects with ISP data collected immediately after injection at Day 1. Some subjects had multiple assessments as FKB327-003 was an extension study following FKB327-002. For systemic analysis of ISP, dataset of interim analysis 2 (IA2) of FKB327-003 OLE study when all patients are completed multiple assessments in the study (at first dose of FKB327 devices or the RP in Period I and II)*. For all other studies (FKB327-001, -002, -004), final dataset are used for the analysis.

* One evaluable VAS data at unscedule visit is included in systemic analysis

3 PLANNED ANALYSES

All summary data will be provided for each treatment group, i.e. FKB327 and Humira®. Unless otherwise specified, categorical data will be summarized using frequencies and percentages. Summary statistics on continuous data will include number of subjects, mean, standard deviation (SD), minimum, median, and maximum.

3.1 Summary of Demographic and Other Baseline Characteristics

Demographic and other baseline characteristics data will be summarized by study. The variables to be summarized include study visit (baseline / week 0 / week 30); device (syringe / auto injector / pre-filled syringe); subject type (healthy volunteer / RA patient); country; gender; age; race; ethnicity; and weight. Age and weight will also be summarized categorically.

3.2 Forest Plot of Injection Site Pain Data

Subgroup analyses will be performed for VAS data. The mean treatment difference and 95%CI will be presented for each subgroup through the forest plot. The subgroups include study, device, subject type, age group, gender, weight group, race, ethnicity, and country.

3.3 Linear Mixed Model

The least squares mean treatment difference and 95%CI will be calculated by using linear mixed model with fixed effects (subject type, age group, gender, race, ethnicity, weight group, device, and treatment) and random effects (country and subject). P-value will also be calculated with the same model.

The other fixed effects will also be estimated with the same model.

3.4 Device

For the VAS data collected after FKB327 injection in all subjects, the device comparison will be performed by using linear mixed model with fixed effects (subject type, age group, gender, race, ethnicity, weight group, and device) and random effects (country and subject). P-value will also be calculated with the same model.

The same model will also be analyzed only for the subjects who experienced three types of FKB327 device.

3.5 Location of Injection

For the VAS data collected after FKB327 injection in all subjects, the location comparison will be performed by using linear mixed model with fixed effects (subject type, age group, gender, race, ethnicity, weight group, device, and location of injection) and random effects (country and subject). P-value will also be calculated with the same model.

3.6 Injection Site Reaction

All injection site reaction data will be summarized by study visit for the randomized subjects with injection site reaction data collected immediately after injection at Day 1. The number and percentage of subjects will be presented for each category of reaction by treatment.

4 REFERENCE

- 1) Puri A, Niewiarowski A, Arai Y, et al., Pharmacokinetics, safety, tolerability and immunogenicity of FKB327, a new biosimilar medicine of adalimumab/Humira, in healthy subjects. Br J Clin Pharmacol. 2017 Jul;83(7):1405-1415.
- 2) Al-Salama ZT. FKB327: An Adalimumab Biosimilar. BioDrugs. 2019 Feb;33(1):113-116.
- 3) NCT02260791: <https://clinicaltrials.gov/ct2/show/NCT02260791> (accessed on 03-Jun-2019)
- 4) NCT02405780: <https://clinicaltrials.gov/ct2/show/NCT02405780> (accessed on 03-Jun-2019)
- 5) Hulio: EPAR – Public assessment report:
https://www.ema.europa.eu/en/documents/assessment-report/hulio-epar-public-assessment-report_en.pdf (accessed on 03-Jun-2019)

5 APPENDICES

Appendix 5-1: FKB327-001 study outline

Appendix 5-2: FKB327-002 study outline

Appendix 5-3: FKB327-003 study outline

Appendix 5-4: FKB327-004 study outline